

Shrey Pant

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EDUCATION

Rajiv Gandhi Institute of Petroleum Technology (RGPT), India

June 2025

B.Tech, Computer Science and Engineering : CGPA: 8.72/10.0

SKILLS

Systems & Internals: Database internals, query execution engines, memory management, concurrency (Tokio, Cats Effect), FFI, compilers

Languages: Rust, Scala, C++, C, Python

Frameworks/Tech: Apache DataFusion, Apache Arrow, Cats Effect, fs2, http4s, FastAPI, Django, Kubernetes, Helm, Docker, Podman, Redis

OPEN SOURCE CONTRIBUTIONS

Apache DataFusion / Apache Arrow-rs | *Query Engine & Columnar Internals (Rust)*

- Designed and shipped a pluggable *SpillFile/TempFileFactory* trait abstraction in DataFusion's spill infrastructure, letting spilling target non-OS backends (Postgres BuffFile, object storage) : [datafusion#21882](#) (merged).
- Migrated Sort-Merge Join's spill-read path from a synchronous state machine to fully async streaming, through several rounds of correctness review with a DataFusion committer : [datafusion#22230](#) (merged).
- Added *with_skip_validation* API parity to Arrow's *StreamDecoder*, unblocking the DataFusion spill-read refactor above : [arrow-rs#9749](#) (merged).

ParadeDB | *Postgres Search & Vectorized Execution (Rust)*

- Built a late-materialization pipeline deferring string/byte dictionary decoding to cut memory/CPU overhead in selective joins and limits : [paradedb#4219](#) (merged), with a follow-up tuning pass : [paradedb#4343](#).
- Implemented dynamic filter pushdown for hash joins into the storage pre-filter layer, preserving SQL three-valued logic with zero-copy Arrow arrays : [paradedb#4306](#) (merged).
- Reworked joinscan memory-pool allocation to match Postgres *work_mem* semantics per-operator-per-partition instead of one shared pool, fixing under-accounted memory budgets for multi-join/sort queries : [paradedb#4413](#).

Cats Effect & Typelevel Ecosystem | *Scala, Scala Native, Functional Concurrency*

- Vilcacora** : designed a type-safe Intermediate Representation and ONNX-to-IR translator, then built the runtime lowering IR to native Cats Effect *IO/Resource* values, integrating LibSVM and MLPack via Scala Native FFI with safe native-memory lifecycle management : github.com/pantShrey/vilcacora. Actively maintained; currently adding BERT/transformer support.
- Diagnosed and contributed a fix toward a WSTP thread-leak causing resource exhaustion in production Scala apps : [cats-effect#4388](#).
- Merged two runtime stability and bug-fix PRs to Cats Effect core : [cats-effect#4352](#) and [cats-effect#4380](#).

HelixDB | *Rust Graph-Vector Database*

- Fixed a CPU core-affinity bug in HelixDB's Tokio runtime integration : [helix-db#659](#) (merged).
- Added first-class Podman support as an alternative container runtime to Docker across the CLI, config, and cross-platform daemon auto-start (macOS/Linux/Windows) : [helix-db#697](#) (merged).

EXPERIENCE

External Contributor, Compensated : ParadeDB

Feb 2026 – Present

ParadeDB (Remote) : ongoing paid open-source work on Postgres search/vectorized execution (see Open Source Contributions above)

Freelance Developer

Oct 2025 – Dec 2025

HelixDB (Remote)

- Engineered a containerized orchestration layer for HelixDB using Rust and Podman, enabling streamlined database lifecycle management.
- Developed the official n8n integration suite, bridging workflows with HelixDB's core API and existing Model Context Protocol (MCP) endpoints.

Google Summer of Code 2025 : Vilcacora (ML Inference in Cats Effect)

June 2025 – Sep 2025

Typelevel (Remote), mentored by Arman Bilge | github.com/pantShrey/vilcacora/tree/handling_bert

- Designed a type-safe Intermediate Representation (IR) and ONNX-to-IR translator with static shape validation and Scala-Native-safe memory allocation, forming the core of an ML inference engine that runs as native Cats Effect *IO/Resource* values.
- Built the runtime module lowering IR to *IO*, integrating LibSVM and MLPack (convolution, pooling, ReLU, Softmax) via Scala Native C++ FFI with *Resource[IO, _]*-managed native memory lifecycles.
- Still actively developing the project post-GSoC : currently extending it with BERT/transformer model support on a working fork.

Software Engineering Intern : Backend & Search Infrastructure

June 2024 – Oct 2024

Plotch (Remote)

- Engineered Khoj, a multimodal search engine with async indexing and multilevel caching, cutting query latency ~5s→≤1s and improving non-English relevance by 35%.
- Architected automated cataloging/ingestion pipelines (RabbitMQ), cutting listing time 95%, indexing time 83%; built a Redis-backed observability stack eliminating \$8k/year in third-party costs.

PROJECTS

Bitcask Implementation | *Rust, Database Internals* : log-structured KV store, O(1) reads, crash recovery via compaction (pantShrey/bitcask_impl)